

Graph Neural Network Modeling of Breast MRI Vascular Networks for Treatment Response Prediction

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Introduction • Motivation

Breast cancer is the most common cancer in women worldwide, and for those diagnosed with locally advanced breast cancer (LABC), neoadjuvant chemotherapy (NAC) is the standard of care. However, only 30% of patients achieve a pathologic complete response (pCR) to NAC, which is associated with improved survival outcomes. NAC is an intensive treatment that can have significant side effects, and patients who do not achieve pCR may require additional treatment. We would like to develop methods to predict which patients will respond to NAC, so that we can better tailor treatment plans and improve outcomes for patients with LABC.

Ultrafast MRI Dataset

The dataset used in this study consists of ultrafast breast MRI scans from $n = 100$ patients with LABC who underwent NAC. The scans were acquired using a 3T MRI scanner and included both pre- and post-contrast images. The dataset also includes clinical information such as age, tumor size, and hormone receptor status, as well as treatment response data (pCR vs. non-pCR).

Graph Representation

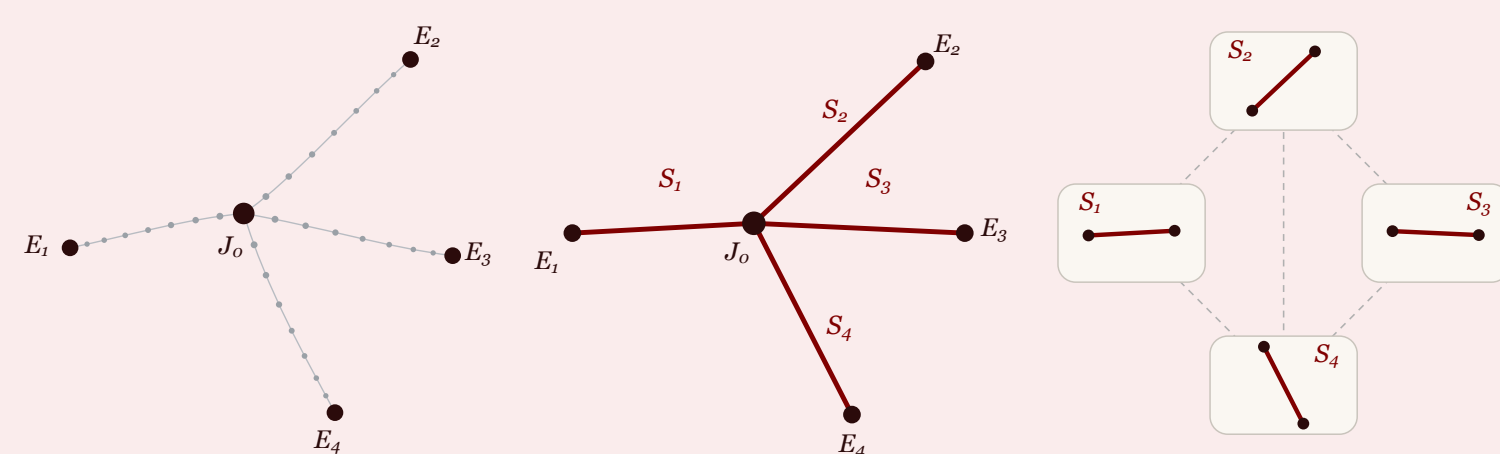


Figure 1. Vascular segments and junctions are converted into graph nodes and edges.

We represent the vascular network of each patient's tumor as a graph, and propose 3 different methods of constructing the graph from the MRI scans. In the following let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph.

- **Voxel-as-node** Let the set of nodes $\mathcal{V}_{\text{vox}}$ be the set of all voxels, and let the set of edges $\mathcal{E}_{\text{vox}}$ be the set of all pairs of adjacent voxels. In this representation, each voxel is a node, and edges connect adjacent voxels.
- **Junction-as-node** Let $\mathcal{V}_{\text{jun}}$ be the set of all junctions (i.e. $e \in \mathcal{E}_{\text{vox}}$ with $\deg(e) > 2$) in the vascular network, and let $\mathcal{E}_{\text{jun}}$ be the set of all pairs of junctions that are connected by a vascular segment.
- **Segment-as-node** Let $\mathcal{V}_{\text{seg}}$ be the set of all vascular segments, and let $\mathcal{E}_{\text{seg}}$ be the set of all pairs of segments that are connected at their endpoints. In this representation, each segment is a node, and edges connect adjacent segments.

Prediction Pipeline



Figure 2. End-to-end prediction pipeline.

Vessel segmentation is performed using a **nnU-Net model**, and voxels are given a binary mask indicating whether they are part of a vessel or not. Then the segmentation is simplified into a one-voxel wide network of vessels using the **tc4d** algorithm. The simplified network is then converted into a graph using one of the three methods described above. Finally, a **Graph Neural Network (GNN)** is trained to predict treatment response from the graph representation of the vascular network.

Graph Neural Network Architecture

After the graph is constructed, we use a GNN to predict treatment response from the graph representation of the vascular network. The GNN consists of several layers of **graph convolutional networks (GCNs)** followed by a global pooling layer and a fully connected layer. The GCN layers learn to aggregate information from neighboring nodes in the graph, while the global pooling layer aggregates information from all nodes in the graph to produce a fixed-size representation of the entire graph. Finally, the fully connected layer maps the graph representation to a binary prediction of treatment response (pCR vs. non-pCR).

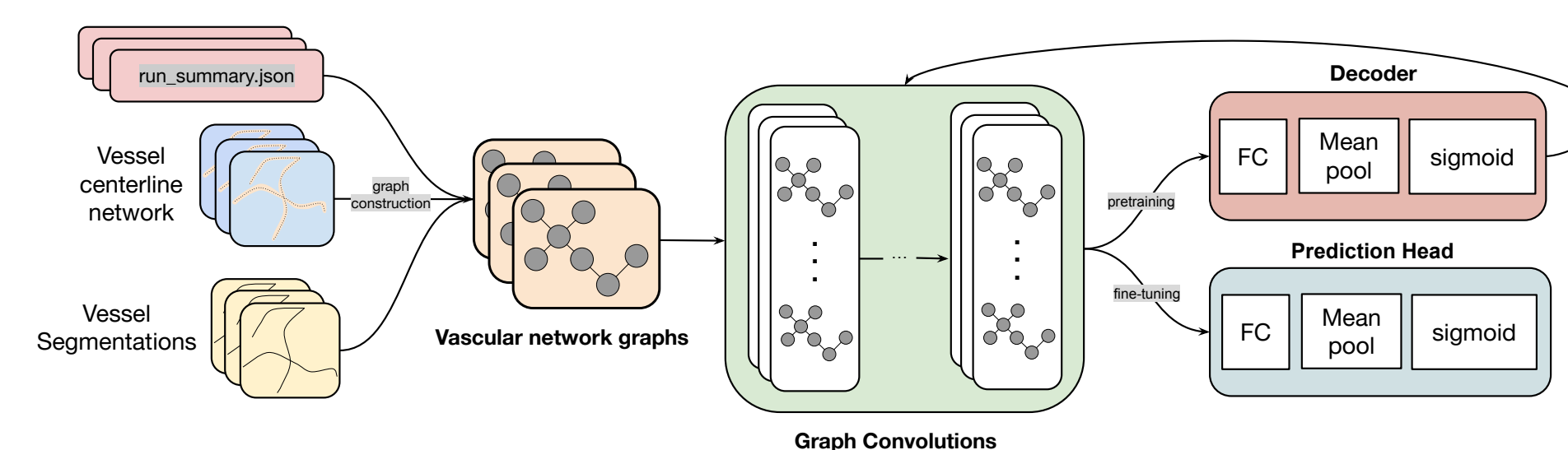


Figure 3. GNN architecture: 4D DCE-MRI through the vessel graph to a pCR probability.

Self-supervised Pretraining

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Results

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Reproducibility

In an effort to promote reproducibility and transparency in our research, we have made our code publicly available on GitHub. The repository includes the code for the vessel segmentation, graph construction, GNN training, and evaluation, as well as instructions for how to reproduce our results.

Conclusion

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References

- [1] Siyi Tang, Jared A. Dunnmon, Khaled Saab, Xuan Zhang, Qianying Huang, Florian Dubost, Daniel L. Rubin, and Christopher Lee-Messer. Self-supervised graph neural networks for improved electroencephalographic seizure analysis, 2022. URL <https://arxiv.org/abs/2104.08336>.